Multi-feature Musical Instrument Sound Classifier

I. Kaminskyj
Electrical & Computer Systems Engineering
Monash University
Clayton Campus, Victoria, Australia.
Email: kaminskyj@eng.monash.edu.au

Abstract

A classification system has been developed which utilises three features to classify isolated monophonic musical instrument sounds: RMS amplitude envelope, constant Q transform (CQT) frequency spectrum and multidimensional scaling analysis (MSA) trajectories. Nineteen musical instruments of definite pitch were used to test the system, covering the note range C3-C6 and representing the major musical instrument families and subfamilies. Musical instrument sound classification was performed using the leave-one-out classification scheme with different implementations of Nearest Neighbour classification (NNC) (varying k values). The classification results were evaluated in terms of classification accuracy and reliability. The individual feature classifier results showed that the MSA trajectory feature provided the best classification results, followed by the CQT frequency spectrum, and then the RMS amplitude envelope. Combining the individual classifiers together to form one classification system provided further improvement, yielding an overall accuracy of 82% and reliability of 83%.

Introduction

Until only very recently, most research in the area of automatic music transcription has focused on the automatic identification of the notes being played by musical instruments, without regard to which instrument played which note (Brown 1992 and Foster et al. 1982). Its really only been in the last 5 or so years where interest has begun to occur in the problem of automatic identification of isolated and/or continuous musical instrument sounds; a problem closely interrelated with that of automatic music transcription.

Kostek and Wieczorkowska (1996) evaluated the use of artificial neural networks (ANN) and rough sets to reliably distinguish isolated musical instruments tones.

For the rough set algorithm, experiments were carried out with sounds from nine orchestral wind instruments. The algorithm input comprised vectors of fourteen features: six steady-state as well as the eight from the attack. Experiments including modified leave-one-out tests yielded an accuracy rate of 65%. For the neural network algorithm, four wind instruments were used for training and testing. Recognition results at 97% were achieved.

Additional experiments (Kostek & Krolikowski 1997) were carried out where training of the ANN was performed using either the left or right channel recordings of four slightly different wind instruments; the McGill University Master Samples (MUMs) CD recordings all being stereo. Classification results for fourteen features ranged from 94.4–100%.

Brown (1998) performed a series of experiments relating to the computer identification of continuous flute, clarinet, oboe and saxophone sounds using autocorrelation and cepstral coefficients as features. She found the best pairwise classification results (90.6 – 100%) were obtained using cepstral coefficients based on a constant Q transform calculated for

short segments of each instrument sound using a k-means algorithm and a Bayes decision rule.

Martin et al. (1998) applied a statistical pattern-recognition technique to the classification of isolated musical instrument tones within a taxonomic hierarchy. Thirty-one perceptually salient acoustic features were extracted from the output of a human auditory model based on the log-lag correlogram. 1023 isolated tones from fifteen orchestral instruments were used to test the system, recorded from the MUMs CDs.

Using 70%/30% splits between training and test data, the classifiers distinguished pizzicato from sustained tones with an accuracy of 99%. Instrument families were identified with an accuracy of 90%, while individual instruments were identified with an accuracy of 70%. Martin et al. claimed that these results compared favourably with human performance on the same task.

Ichiro (2000) designed a computer-based classifier to recognise simple continuous tones from the MUMs CDs of musical instrument sounds. 39 different timbres from 23 orchestral instruments played at different pitches were used as samples for an exemplar-based learning system that incorporated a k-NNC with genetic algorithm.

The weighted features used during classification included moments calculated from both the steady-state and dynamically changing spectral envelopes as well as spectral irregularity and tristimulus features describing the spectral envelopes. While the recognition accuracy varied greatly between the instruments, the average overall accuracy was 68%.

Although the results obtained previously are encouraging, they are generally limited to a restricted number and type of musical instruments. In some cases, only wind instruments are used, while in others, only orchestral instruments. The classification system described in this paper uses a more general selection of instruments together with a novel approach to combine individual feature classifiers, applicable particularly where limited data sets are available. The combination process makes use of the application of weighted ranked individual classifier results together with the confusion matrix as a mechanism for determining the relative confidence that can be applied to each classifier classification result. The classification results obtained for nineteen musical instrument show the merit of the approach taken.

Data collection

Sounds from nineteen musical instruments of definite pitch were chosen for system testing. 517 recordings of these instruments were taken directly from the MUMS CDs. The instruments included: guitar (plucked string), violin, cello and double bass (bowed string), piano (struck string), flute (air reed wind), accordion, clarinet, saxophone (single mechanical reed wind), oboe and bassoon (double mechanical reed wind), organ (air/mechanical reed wind), trumpet, trombone, French horn, and tuba (lip reed wind) and xylophone, glockenspiel and marimba (percussive definite pitch) (Olson 1967).

The note range used was C3-C6 of the equally tempered musical scale. A three-octave note range was chosen to limit the system complexity, to provide practical limits on the number of recordings required and to simplify instrument selection. Not all instruments covered the complete note range, but each had at least some notes falling within it. Recordings were made from the MUMs CDs using the SoundBlaster WaveStudio program that allowed recording, display, editing and playback of sounds. All recordings comprised 2 seconds of edited sound.

Data preprocessing (feature extraction)

Once the raw musical instrument recordings were edited, they needed to be preprocessed before classification. The aims of this stage of the system were (a) to reduce the amount of data that needed to be handled by the classifier(s), as well as (b) to perform feature extraction, so as to make the classification task as easy as possible. The three features discussed below were evaluated for their effectiveness in instrument classification. They were chosen either because they had been used successfully in previous research in solving similar problems (Kaminskyj 1995) or had proven important in allowing humans to discern between musical instrument sounds (Handel 1995) or simply looked intuitively promising (Hourdin et al.1997).

RMS amplitude envelope

Many schemes exist for measuring the amplitude envelope of a waveform. The approach used in this research involves the calculation of the short-term RMS value over 2-3 periods of the fundamental frequency (Kaminskyj 1996).

CQT frequency spectrum

Many spectral representations are possible for musical waveforms. The representation used in this research is the constant Q transform developed by Brown (1991), because of its logarithmically spaced spectral bins which align with musical scale note frequencies.

MSA trajectories

MSA was performed on harmonic extracts of CQT spectra of musical instrument sounds (Kaminskyj 1999). Principal component analysis (PCA) (Jollife 1986) was applied to the resultant CQT spectral snapshots to determine the most significant attributes characterising each sound. The resultant sound trajectories yielded a "tone signature" for each instrument sound.

Classification algorithms

For "curse of dimensionality reasons" (Pandya & Macy 1996), the classification approach adopted for this work uses three *k*-NNCs utilising the Euclidean distance measure to classify musical instruments using different features and the leave-one-out classification scheme (Duda & Hart 1973).

With this scheme, all recordings were used to test the classifiers, but only one at a time. As each recording was classified, the remaining recordings formed labelled reference samples to be used to compare against the test recording. In this way, all recordings were available to evaluate the classification system performance. In addition, a large number of labelled reference samples were available to aid test sample comparison and classification.

With the leave-one-out scheme, classification was obtained by determining the *k* closest reference samples to the test sample, with *k* values of 1, 3 and 5 explored. Either weighted or unweighted majorities were then used to determine the most likely instrument to have produced the test sample. Classification of instruments sounds occurred using either the raw feature waveforms (Direct) or alternatively, after PCA was performed on these waveforms.

Classifier results were evaluated in terms of accuracy and reliability (Pandya & Macy 1996). Classification accuracy indicates how many of the test samples were correctly classified. Classification reliability on the other hand, provides an indication as to the confidence that can be placed on the classifier results; in other words, if a classifier gives a clarinet result for example, how often is this result correct. A classifier with good classification accuracy but poor reliability for certain instruments is of limited use for these specific instruments.

Initially, the classification system was devised such that a test sample was compared against all reference samples. This approach however resulted in misclassifications which were obviously incorrect (eg. C3 marimba tone misclassified as a glockenspiel, whose note range doesn't extend below G6). As it has been assumed that the note of the test sample is known *a priori* by the classification system, such an obvious misclassification had to be somehow avoidable.

As an end to avoiding such misclassifications, octave search limits were then applied when performing comparisons between the test and reference samples for each of the three feature classifiers. Octave search limits invariably improved the classification results obtained. For the most promising input feature type (direct or PCA) for each individual classifier, the search limits were then further reduced to ± 5 notes, ± 4 notes, ± 3 notes (half octave), ± 2 notes and ± 1 notes. In this way, the optimum search limits for each individual feature classifier was determined. For example, a ± 2 note limit applied to a G3 test tone meant that comparisons for this test tone only occurred with F3, F3#, G3, G3# and A3 reference tones.

Results

Tables 1-6 show the classification accuracy results for the individual feature classifiers. Table 7 shows the results obtained when combining the individual feature classifier results in a variety of ways, including using the confusion matrix (Pandya & Macy 1996) instrument accuracy (Acc) and reliability (Rel) values as combination weights (See Tables 8-10).

Table 1: RMSPCA

NNC k Value	Limits	% accuracy Majority (unwghtd)	% accuracy Majority (weighted)
1	none	39%	N/A
3	none	41%	39%
5	none	43%	43%
1	octave	46%	N/A
3	octave	45%	46%
5	octave	48%	48%

Table 2: RMS Direct

NNC k Value	Limits	% accuracy Majority (unwghtd)	% accuracy Majority (Weighted)
1	none	48%	N/A
3	none	50%	48%
5	none	49%	49%
1	octave	56%	N/A
3	octave	58%	56%
5	octave	56%	57%
1	± 5	56%	N/A
3	± 5	60% (best)	56%
5	± 5	55%	59%
1	± 4	56%	N/A
3	± 4	59%	56%
5	± 4	55%	58%
1	± 3	57%	N/A
3	± 3	56%	57%
5	± 3	53%	56%
1	± 2	54%	N/A
3	± 2	54%	54%
5	± 2	48%	52%
1	± 1	46%	N/A
3	± 1	41%	46%
5	± 1	30%	38%

Tables 8-10 show the confusion matrices for the best individual feature classifiers in Tables 1-6, while Table 11 shows the confusion matrices for the best combined feature classifier in Table 7.

Discussion

The results in Tables 1-6 indicate that the best classification results for the RMS amplitude envelope NNC were obtained using the direct waveform feature (accuracy 60% & reliability 60%). On the other hand, for the CQT spectral NNC and MSA trajectory NNC, best results were obtained with the PCA features (CQT accuracy 66% & reliability 68%; MSA accuracy 75% & reliability 76%). It is perhaps not surprising that the latter two feature NNCs both favour the PCA features as both are closely related together; the MSA trajectories being derived from the CQT spectral snapshots over the evolution of each musical instrument sound.

Of the three features, the MSA trajectories provide the most powerful discrimination for the nineteen instruments used for testing. This is perhaps not too surprising as these trajectories provide the most comprehensive amount of information regarding both the temporal and spectral characteristics of each individual instrument sound.

Table 3: CQTPCA

NNC k Value	Limits	% accuracy Majority (unwghtd)	% accuracy Majority (weighted)
1	none	62%	N/A
3	none	58%	62%
5	none	58%	60%
1	octave	65%	N/A
3	octave	62%	65%
5	octave	61%	63%
1	± 5	65%	N/A
3	± 5	63%	65%
5	± 5	62%	63%
1	± 4	66%	N/A
3	± 4	64%	66%
5	± 4	62%	63%
1	± 3	66% (best)	N/A
3	± 3	64%	66%
5	± 3	61%	63%
1	± 2	65%	N/A
3	± 2	62%	65%
5	± 2	57%	62%
1	± 1	61%	N/A
3	± 1	55%	61%
5	± 1	45%	55%

Table 4: CQT Direct

NNC k Value	Limits	% accuracy Majority (unwghtd)	% accuracy Majority (Weighted)
1	none	51%	N/A
3	none	50%	51%
5	none	49%	51%
1	octave	53%	N/A
3	octave	53%	53%
5	octave	52%	53%

Table 5: MSAPCA (normalised trajectory)

NNC k Value	Limits	% accuracy Majority (unwghtd)	% accuracy Majority (weighted)
1	none	67%	N/A
3	none	68%	67%
5	none	66%	68%
1	octave	71%	N/A
3	octave	71%	71%
5	octave	71%	72%
1	± 5	72%	N/A
3	± 5	72%	72%
5	± 5	70%	73%
1	± 4	74%	N/A
3	± 4	74%	74%
5	± 4	71%	72%
1	± 3	74%	N/A
3	± 3	75% (best)	74%
5	± 3	70%	73%
1	± 2	73%	N/A
3	± 2	70%	73%
5	± 2	67%	70%
1	± 1	70%	N/A
3	± 1	62%	70%
5	± 1	49%	59%

Table 6: MSA Direct

NNC k Value	Limits	% accuracy Majority (unwghtd)	% accuracy Majority (weighted)
1	none	53%	N/A
3	none	51%	53%
5	none	51%	51%
1	octave	56%	N/A
3	octave	55%	56%
5	octave	55%	55%

Table 7: Combined Classifier

NNC k Value	Confn Matrix Wghtg Method	% accuracy Majority (unwghtd)	% accuracy Majority (weighted)
1	none	78%	N/A
3	none	80%	80%
5	none	75%	81%
1	Rel	78%	N/A
3	Rel	81%	81%
5	Rel	77%	82% (best)
1	Acc	78%	N/A
3	Acc	80%	78%
5	Acc	75%	81%
1	Rel+Acc	78%	N/A
3	Rel+Acc	80%	79%
5	Rel+Acc	74%	79%

The varying search limit study results of Tables 2, 3 and 5, indicate that for the RMS amplitude envelope feature, best results are obtained for a roughly octave search limit.

For the CQT and MSA features however, best results are obtained for a half octave search limit, before extraneous confusions with reference samples further away in frequency became a problem. These results

tend to highlight that the proximity effect (the range of notes over which features remain roughly invariant and instrument confusions minimal) is feature dependent.

The combined feature results highlight that the confusion matrix reliability weighting provides the best classification results [*k=5*, weighted majority] (accuracy 82% and reliability 83%), although this is only marginally better than the results obtained for the non-confusion matrix weighted results. As reliability indicates the confidence that can be placed on any given classifier results, it is not surprising that of the three confusion matrix weightings, it provides the best results. What is surprising is that such little improvement was obtained by using the confusion matrix weightings. This leads to the conclusion that the rankings of the individual classifiers beyond the first choice provide very useful information when combining the individual classifier results together.

For the combined feature classifier, only two instruments of the nineteen tested did not provide reasonable classification results (deemed a minimum of 67% classification accuracy and reliability). These two instruments were cello and saxophone, although even cello came very close (accuracy 65% & reliability 100%). This result indicates that for these two instruments at least, the three features chosen were not adequate enough to allow reliable discernment.

It is interesting to note that few researchers have paid any attention to the issue of reliability of their classification results; most focused purely on the overall system accuracy.

Table 8: CONFUSION MATRIX RMS Direct: ±5 NOTE LIMIT k=3 UNWEIGHTED MAJORITY

	Acc	**Bas**	**Cel**	**Cla**	**Dbb**	**Flu**	**Glo**	**Gui**	**Hor**	**Mar**	**Obo**	**Org**	**Pia**	**Sax**	**Tro**	**Tru**	**Tub**	**Vio**	**Xyl**	%Acc
Acc	22		1	1	2	3									2	1		2		65%
Bas		12		1					2					1	1		2			63%
Cel	1		5	1				2	3			5			1		2			25%
Cla	1	1		30										4						83%
Dbb	1				11							2						3		65%
Flu	6					16					1	1				1				64%
Glo							6													100%
Gui			2					33					1						1	89%
Hor	1			1					21			1		1	2					78%
Mar							2			23			5						7	62%
Obo	1	1									17	1		1	3	1	1	1		63%
Org	2	4	4	1	1	4			1		1	12		1	1	2		3		32%
Pia								3		11			20						3	54%
Sax		1		13										22	1					59%
Tro		2		1	1				2			1			17	3	1			61%
Tru	3														7	13		2		52%
Tub		3		1					1		2				4		9			45%
Vio	2				6	1					1				2	3	2	13		43%
Xyl								1		5			4						10	50%
%Rel	55	50	42	60	52	67	75	85	70	59	77	52	67	73	41	54	53	54	48	60/60

Table 9: CONFUSION MATRIX CQTPCA: ± 3 NOTE LIMIT k=1NNC

	Acc	Bas	Cel	Cla	Dbb	Flu	Glo	Gui	Hor	Mar	Obo	Org	Pia	Sax	Tro	Tru	Tub	Vio	Xyl	%Acc
Acc	22		1		1	1						2	1	1	2	2		1		65%
Bas		12				1			2				2				2			63%
Cel	1		10	1	7					1										50%
Cla	1		2	22							1	1		4	1	3		1		61%
Dbb			1	2	8			2						2	1		1			47%
Flu					1	21			1		1				1					84%
Glo							5											1		83%
Gui	1				1	1		28			2		2	1	1					76%
Hor									21						6					78%
Mar								2		28		1	1				2		3	76%
Obo				5		1		4			10	1		3	1	1		1		37%
Org	5		1			1						26		1		3				70%
Pia								4	2				28	1					2	76%
Sax	2	1		2	1			3	2		5		1	15	2	1		2		41%
Tro									8					1	19					68%
Tru	1			2							1	1				19		1		76%
Tub		2							1								17			85%
Vio	3		3	1							2	2		2		1		16		53%
Xyl								1					2	2					15	75%
%Rel	61	80	56	63	42	81	100	64	57	97	45	76	76	45	56	63	77	70	75	68/66

Although in general, overall reliability tracks overall accuracy quite closely, this is not necessarily the case for an individual instrument. As reliability directly indicates the confidence that can be placed on the results of a classifier for a given instrument, it is of primary importance in quantifying the performance of a classifier; accuracy may simply not be enough.

For example, consider clarinet in Table 8. For this classifier, clarinet was accurately classified 83% of the time, which at first glance is encouraging. Unfortunately, when this classifier provided a clarinet result, it was only right 60% of the time; considerably worse than one would expect from just the accuracy figure.

Table 10: CONFUSION MATRIX MSAPCA: ±3 NOTE LIMIT k=3 UNWEIGHTED MAJORITY

	Acc	Bas	Cel	Cla	Dbb	Flu	Glo	Gui	Hor	Mar	Obo	Org	Pia	Sax	Tro	Tru	Tub	Vio	Xyl	%Acc
Acc	28					1									2	2		1		82%
Bas		14				1			2						1		1			74%
Cel			12	2	3			1								1	1			60%
Cla				25		1					1			8		1				69%
Dbb					15	1												1		88%
Flu	1				1	13		1	3					1	4			1		52%
Glo							6													100%
Gui			2					32					2				1			86%
Hor									24						3					89%
Mar										31			2						4	84%
Obo				3							13	1		3	1	4		2		48%
Org				1								31				5				84%
Pia								5		5			25						2	68%
Sax	1			9		2		2			3			17	2	1				46%
Tro									6						22					79%
Tru											3					22				88%
Tub																	20			100%
Vio	1				2	1					3				2			21		70%
Xyl							1			2			1						16	80%
%Rel	90	100	86	63	71	65	86	78	69	82	57	97	83	59	59	61	87	81	73	76/75

Table 11: CONFUSION MATRIX NNC Combined: k=5 WEIGHTED MAJORITY, Confusion Matrix weighted (reliability)

	Acc	Bas	Cel	Cla	Dbb	Flu	Glo	Gui	Hor	Mar	Obo	Org	Pia	Sax	Tro	Tru	Tub	Vio	Xyl	%Acc
Acc	27					3						2			1	1				79%
Bas		16				1											2			84%
Cel			13	1	2			2								1	1			65%
Cla				31		1					1			3						86%
Dbb					14							1			1			1		82%
Flu					1	22			1						1					88%
Glo							6													100%
Gui								35					1				1			95%
Hor									25						2					93%
Mar							1			33			1						2	89%
Obo	1			3							18	1		1		1		2		67%
Org												33				4				89%
Pia								3		2			31						1	84%
Sax		1		10		1		1	2					21		1				57%
Tro									7						21					75%
Tru											1					24				96%
Tub																	20			100%
Vio	2				3						1	1			1			22		73%
Xyl								1		3			1						15	75%
%Rel	90	94	100	69	70	79	86	83	71	87	86	87	91	84	78	75	83	88	83	83/82

Similar results are obtained for bassoon in Table 8, French horn in Table 9 and trombone in Table 10.

Conclusion

The classification results obtained for the combined classifier system provide encouragement that the approach taken bears merit, particularly where limited data sets are available. This has been achieved using a comprehensive, yet general selection of instruments (rather than just wind or orchestral), which should in turn make the results obtained more generally applicable across a wide range of instruments. It is also surprising that the results were obtained for so many instrument using only three features. This shows in particular, the strength of the MSA trajectory feature for discriminating between different musical instruments.

The results achieved question the validity of modeling a musical instrument classification system exclusively on the capabilities of the human auditory system. Given the limitations humans have demonstrated in identifying musical sounds, such approaches necessarily restrict system performance to at best mimic the results achieved by humans for such tasks; yet never possibly surpass them.

Although the system described is principally intended for off-line analysis of musical instruments sounds, Ichiro (2000) recently demonstrated that using similar classification algorithms with current day computing facilities, real-time performance is achievable.

The work described is but one more step towards the development of a robust musical instrument classifier. Much work still remains. Possible directions for future work include exploring (a) new features, eg. spectral centroid and cepstral coefficients (b) the generalisation capabilities of the combined classifier (isolated and continuous tones) and (c) extending the system to handle polyphonic sounds.

Bibliography

Brown, J. C. “Calculation of a Constant Q Spectral Transform”. J. Acoust. Soc. Am., 89 (1) 1991, pp. 425-34.

Brown, J. C., "Musical Fundamental Frequency Tracking using Pattern

Recognition Method", J. Acoust. Soc. Am., 90 (3) 1992, pp. 1394-1402
Brown, J. C., "Musical instrument identification using autocorrelation coefficients", Proc. Int. Symp. Musical Acoustics, Washington, 1998, pp. 291-5.
Duda, R. O., and Hart, P. E., "Pattern Classification and Scene Analysis", Wiley-Interscience, 1973
Foster, S., et al "Toward an Intelligent Editor of Digital Audio: Signal Processing Methods", Computer Music Journal, 6 (1), 1982 pp. 42-51
Fujinaga, I., and MacMillan, K., "Realtime recognition of orchestral instruments", accepted for Proc. Int. Comp. Music Conf. 2000.
Handel, S., "Timbre perception and auditory object identification", in *Hearing*, Moore, B. C. J., New York: Academic Press, 1995.
Hourdin, C., Charbonneau, G., et al. "A Multidimensional Scaling Analysis of Musical Instruments' Time-Varying Spectra", Computer Music Journal, 21/2, 1997, 40-55.
Jollife, I. T., "Principal Component Analysis", Springer-Verlag, 1986.
Kaminskyj, I., "Multidimensional Scaling Analysis of Musical Instrument Sounds' Spectra", Proc. ACMA Conf. 1999 Wellington, NZ, pp. 36-39.
Kaminskyj, I., and Materka, A., "Automatic Monophonic Musical Instrument Sound Identification System", Proc. ACMA Conf 1996, Brisbane, Aust., pp. 41-46.
Kaminskyj, I., and Materka, A., "Automatic Source Identification of Monophonic Musical Instrument Sounds", *IEEE Int. Conf. Neural Networks 1995*, Vol. 1, Perth, WA, pp. 189-194.
Kostek, B., and Wieczorkowska, A., "Study of Parameter Relations in Musical Instrument Patterns", Proc. 100th Audio Eng. Soc. Conv., Preprint 4173, Copenhagen, 1996.
Kostek, B., and Krolikowski, R., "Application of Artificial Neural Networks to the Recognition of Musical Sounds", Archives of Acoustics, 22 (1), pp. 27-50, Poland, 1997.
Martin, K. D., and Kim, Y. E., "Musical instrument identification: A pattern-recognition approach", 136th Meeting J. Acoust. Soc. Am., Oct. 1998.
Olson, H. F., "Music, Physics and Engineering", 2nd Ed, Dover Publications, 1967.
Pandya, A. S., and Macy, R. B., "Pattern Recognition with Neural Networks in C++", CRC Press, 1996.